Modern React 18 & TypeScript Masterclass

Lesson 1: Course Introduction & Architecture Setup

Instructor: A. Sharma • Category: Frontend Engineering • Duration: 25 Mins

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1. Overview & Learning Objectives

Welcome to Lesson 1 of the Modern React 18 & TypeScript Masterclass. This lecture establishes foundational principles for writing production-grade React code in modern enterprise environments.

- **Modern Toolchain Setup:** Understand Vite 5 compilation, ES modules, and TypeScript compiler configurations.
- **React 18 Concurrent Rendering:** Deep dive into fiber reconciler improvements, automatic batching, and priority lanes.
- **Strict Type Safety:** Learn how generic props, discriminated unions, and ReactNode definitions eliminate runtime faults.
- **Production Project Structure:** Build scalable feature-folder architectures designed for large cross-functional engineering teams.

2. React 18 Core Architectural Pillars

React 18 introduces concurrent rendering foundations that allow React to interrupt, pause, resume, and abandon renders without blocking the browser main thread.

Feature	Legacy React 17	Modern React 18+
Batching	Synthetic events only	Automatic everywhere (Promises, timeouts, native handlers)
Rendering	Synchronous & blocking	Interruptible concurrent fiber execution
Transitions	Manual debouncing required	Native startTransition & useTransition primitives
SSR Streaming	All-or-nothing HTML render	Progressive Suspense selective hydration

3. Standard TypeScript Starter Boilerplate

To kick off this lesson hands-on exercise, initialize your local workspace using Vite with the React-TS template:

```
npm create vite@latest edu-react-app -- --template react-ts
cd edu-react-app
npm install
npm run dev
```

Verify your tsconfig.json has "strict": true and "jsx": "react-jsx" enabled before proceeding to Module 2: State Primitives and Hooks.

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